

Planar Cancellation and Energy Estimates for Heat-Conducting Compressible Flow

Abstract

We study cancellation and energy estimates relevant to the two-dimensional full compressible Navier–Stokes system with vacuum at infinity. We prove that an integrable distributional divergence of a square-integrable vector field on the plane has zero integral, by constructing cutoffs with scale-invariant gradient energy and controlling their flux tails. For smooth rapidly decaying solutions of the full system, we derive the kinetic and internal energy balances, the pressure equation with its thermal sources, and an estimate for the square of the total specific energy. These identities exhibit useful cancellations and the terms requiring further estimates: the pressure-square integral, temperature-weighted viscous heating, and the velocity-weighted energy flux. The original local strong-solution class is recorded separately from the additional hypotheses used to justify these computations. The estimates do not bound the continuation quantities in that class, and no small-energy global-existence result is established.

Keywords Compressible Navier–Stokes equations; vacuum; planar cancellation; heat conduction; energy estimates

1 Introduction

Consider the full compressible Navier–Stokes equations on $\mathbb{R}^2$:

$$\rho_t + \operatorname{div}(\rho u) = 0, \quad (1.1)$$

$$(\rho u)_t + \operatorname{div}(\rho u \otimes u) - \mu \Delta u - (\mu + \lambda) \nabla \operatorname{div} u + \nabla P = 0, \quad (1.2)$$

$$c_v((\rho \theta)_t + \operatorname{div}(\rho \theta u)) - \kappa \Delta \theta + P \operatorname{div} u = Q(u), \quad (1.3)$$

$$P = R \rho \theta, \quad Q(u) = 2\mu |D(u)|^2 + \lambda (\operatorname{div} u)^2. \quad (1.4)$$

Here $\rho \geq 0$, $u \in \mathbb{R}^2$, and $\theta \geq 0$ denote density, velocity, and temperature, respectively, and

$$D(u) = \frac{\nabla u + (\nabla u)^{\operatorname{tr}}}{2}, \quad \mu > 0, \quad \mu + \lambda \geq 0, \quad R, c_v, \kappa > 0. \quad (1.5)$$

The convention is $(\nabla u)_{ij} = \partial_j u_i$. The initial condition is prescribed in conservative variables:

$$(\rho, \rho u, \rho \theta)(x, 0) = (\rho_0, \rho_0 u_0, \rho_0 \theta_0)(x). \quad (1.6)$$

The far-field condition in the local theory of Wang [1] is $(\rho, u, \theta)(x, t) \rightarrow (0, 0, 0)$ as $|x| \rightarrow \infty$, $t > 0$, with the qualification “in some weak sense”. We retain that qualification; it is not replaced here by pointwise decay or by an independently chosen normalization of u or θ .

The initial data in [1, Theorem 1.1] satisfy, for $a > 1$, $q > 2$, and $\eta_0 > 0$,

$$\bar{x} = (e + |x|^2)^{1/2} \log^{1+\eta_0}(e + |x|^2), \quad (1.7)$$

and

$$0 \leq \bar{x}^a \rho_0 \in L^1 \cap H^1 \cap W^{1,q}, \quad \theta_0 \geq 0, \quad (1.8)$$

$$\sqrt{\rho_0} u_0, \sqrt{\rho_0} \theta_0 \in L^2, \quad \nabla u_0 \in H^1, \quad \nabla \theta_0 \in L^2, \quad (1.9)$$

$$-\mu \Delta u_0 - (\mu + \lambda) \nabla \operatorname{div} u_0 + R \nabla(\rho_0 \theta_0) = \sqrt{\rho_0} g, \quad g \in L^2. \quad (1.10)$$

All unqualified function spaces below are over $\mathbb{R}^2$. In particular, (1.9) requires $\int \rho_0 \theta_0^2 dx < \infty$. It does not separately require $\theta_0 \in L^2$ or $u_0 \in L^2$. There is no second thermal compatibility condition in these printed hypotheses.

Put

$$\alpha = \min\{a/2, 1\}, \quad p_u = \frac{4}{\alpha}, \quad p_\theta = \frac{6}{2\alpha - 1}, \quad q_1 \in (2, \infty). \quad (1.11)$$

Thus $1/2 < \alpha \leq 1$ and both p_u, p_θ are finite. On a local existence interval $[0, T_0]$, the class stated in [1, (1.8)–(1.9)] is

$$\begin{aligned} \rho &\in C([0, T_0]; L^1 \cap H^1 \cap W^{1,q}), \\ \bar{x}^a \rho &\in L^\infty(0, T_0; L^1 \cap H^1 \cap W^{1,q}), \end{aligned} \quad (1.12)$$

$$\nabla u \in L^\infty(0, T_0; H^1) \cap L^2(0, T_0; W^{1,q}), \quad u \in L^\infty(0, T_0; L^{p_u}), \quad (1.13)$$

$$\sqrt{\rho} u, \sqrt{\rho} \theta, \sqrt{\rho} u_t, \nabla \theta, \sqrt{t} \sqrt{\rho} \theta_t, \sqrt{t} \nabla^2 \theta, \bar{x}^{\alpha/2} \nabla u \in L^\infty(0, T_0; L^2), \quad (1.14)$$

$$\nabla u_t, \sqrt{\rho} \theta_t, \nabla^2 \theta, \sqrt{t} \nabla \theta_t \in L^2(\mathbb{R}^2 \times (0, T_0)), \quad (1.15)$$

$$\theta \in L^2(0, T_0; L^{p_\theta}), \quad \sqrt{t} \nabla^2 \theta \in L^2(0, T_0; L^{q_1}). \quad (1.16)$$

The strong-solution definition in that source requires the derivatives in the equations to be regular distributions and the equations to hold almost everywhere. The condition in (1.13) concerns u itself, not a class of velocities modulo constants.

Let $T_{\max}$ denote the maximal existence time in this class. The continuation criterion stated in [1, Theorem 1.2] is

$$T_{\max} < \infty \implies \lim_{T \uparrow T_{\max}} \int_0^T (\|\operatorname{div} u\|_\infty + \|u\|_{p_u}) dt = \infty. \quad (1.17)$$

Both time exponents in this criterion are one. A bound on compact subintervals whose constants may diverge as $T \uparrow T_{\max}$ does not establish the required continuation bound.

The small quantity motivating this investigation is the physical initial energy

$$E_0 = \int_{\mathbb{R}^2} \left(\frac{1}{2} \rho_0 |u_0|^2 + c_v \rho_0 \theta_0 \right) dx. \quad (1.18)$$

It is finite under (1.8)–(1.9): with $m_0 = \int \rho_0 dx$, Cauchy–Schwarz gives $\int \rho_0 \theta_0 dx \leq m_0^{1/2} \|\sqrt{\rho_0} \theta_0\|_2$. The question is whether sufficiently small E_0 , for prescribed finite bounds on the other initial quantities, ensures global continuation in (1.12)–(1.16). No small higher norm, small mass, zero total momentum, second moment, positive density lower bound, or support restriction is appended to that question. No mass normalization is used below. The printed initial hypotheses in [1] do not separately

state $m_0 > 0$; an argument using a positive local mass must specify that premise and the dependence of its constants. Our cancellation theorem uses no mass condition.

The planar cutoff argument behind the cancellation methods of Fan–Lü–Wang [3] concerns integrable divergences of square-integrable fields. Section 2 proves that general result without a total-momentum assumption. The energy approach, for which Li–Xin [2] is a useful reference in the compressible setting, must retain pressure work, viscous heating, and thermal conduction in (1.3). Sections 3–5 derive corresponding estimates with their hypotheses displayed. They do not establish the missing uniform bounds in (1.17).

Throughout, $\|\cdot\|_p$ denotes the $L^p(\mathbb{R}^2)$ norm, and $\dot{f} = f_t + u \cdot \nabla f$ denotes the material derivative. Unless otherwise specified, constants C depend only on the physical coefficients in (1.5); C_ε may additionally depend on the positive parameter ε .

2 A planar cancellation theorem

We first isolate the limiting argument. Its hypotheses concern actual Lebesgue integrals and do not include a differential equation.

Lemma 2.1 *Let $f, h \in L^1(\mathbb{R}^2)$ be real-valued, with $h \geq 0$ almost everywhere. Let χ_n be measurable real functions such that $|\chi_n(x)| \leq B$ for every $n \in \mathbb{N}$ and $x \in \mathbb{R}^2$, for a fixed $B \geq 0$, and $\chi_n \rightarrow 1$ almost everywhere. Suppose that, for a fixed $C \geq 0$,*

$$\left| \int f \chi_n dx \right| \leq C \left(\int_{\{|x| \geq n\}} h dx \right)^{1/2} \quad (n \in \mathbb{N}). \quad (2.1)$$

Then $\int f dx = 0$.

Proof The products $f \chi_n$ are integrable because $|f \chi_n| \leq B|f|$. For each fixed x , the indicator of $\{|x| \geq n\}$ tends to zero. The functions $h \mathbf{1}_{\{|x| \geq n\}}$ are dominated by h , so dominated convergence gives

$$\lim_{n \rightarrow \infty} \int_{\{|x| \geq n\}} h dx = 0.$$

These integrals are nonnegative and their square roots also tend to zero. On the other hand, dominated convergence with dominating function $B|f|$ yields

$$\lim_{n \rightarrow \infty} \int f \chi_n dx = \int f dx.$$

Taking limits in (2.1) proves the conclusion. $\square$

Theorem 2.2 *Let $f \in L^1(\mathbb{R}^2; \mathbb{R})$ and $V \in L^2(\mathbb{R}^2; \mathbb{R}^2)$. Suppose that $f = \operatorname{div} V$ in distributions, that is,*

$$\int f \varphi dx = - \int V \cdot \nabla \varphi dx \quad \text{for every } \varphi \in C_c^\infty(\mathbb{R}^2). \quad (2.2)$$

Then $\int_{\mathbb{R}^2} f dx = 0$.

Proof Construct a cutoff by setting

$$b(t) = \begin{cases} \exp(-1/t), & t > 0, \\ 0, & t \leq 0, \end{cases} \quad \eta(x) = \frac{b(4 - |x|^2)}{b(4 - |x|^2) + b(|x|^2 - 1)}.$$

Every derivative of b from the right vanishes at zero, so b is smooth. The two arguments in the denominator cannot both be nonpositive. Thus η is smooth, $0 \leq \eta \leq 1$, $\eta = 1$ on $|x| \leq 1$, and $\eta = 0$ on $|x| \geq 2$. In particular, $K_\eta = \int |\nabla \eta|^2 dx < \infty$. For $L > 0$, define $\chi_L(x) = \eta(x/L)$. The chain rule gives

$$\nabla \chi_L(x) = L^{-1} \nabla \eta(x/L), \quad \text{supp } \nabla \chi_L \subset \{L \leq |x| \leq 2L\}. \quad (2.3)$$

The planar change of variables $x = Ly$ yields

$$\int |\nabla \chi_L(x)|^2 dx = L^{-2} L^2 \int |\nabla \eta(y)|^2 dy = K_\eta. \quad (2.4)$$

All integrals here are finite by smoothness and compact support.

Take $L = n + 1$. The actual test function χ_{n+1} is admissible in (2.2), so

$$\begin{aligned} \left| \int f \chi_{n+1} dx \right| &\leq \int_{\{|x| \geq n\}} |V| |\nabla \chi_{n+1}| dx \\ &\leq K_\eta^{1/2} \left(\int_{\{|x| \geq n\}} |V|^2 dx \right)^{1/2}. \end{aligned} \quad (2.5)$$

The tested forcing is integrable since $|\chi_{n+1}| \leq 1$. The flux pairing is integrable by Cauchy–Schwarz, using $V \in L^2$ and (2.4). Moreover, $\chi_{n+1}(x) = 1$ for all sufficiently large n at every fixed x . Lemma 2.1, with $h = |V|^2$, $B = 1$, and $C = K_\eta^{1/2}$, proves the conclusion. $\square$

The scale-independent bound (2.4) is the planar feature of this argument. In dimension d , the same dilation gives $L^{d-2} K_\eta$. No L^1 assumption on V was used in the plane, and the constant function one was never used as a distributional test.

Theorem 2.2 is a statement about f and V with the displayed function-space hypotheses. It is not an assertion that a particular forcing or stress of (1.1)–(1.3) satisfies them. Such an application requires its own verification of the distributional identity, L^1 integrability of the forcing, and L^2 integrability of the flux. The theorem alone supplies neither weighted elliptic bounds nor the norms in (1.17).

3 Physical energy and pressure work

For Sections 3–5, impose the following sufficient regularity condition explicitly: (ρ, u, θ) is a smooth solution with $\rho, \theta \geq 0$, and these functions and all spatial and time derivatives used below decay faster than every inverse power of $|x|$, uniformly on compact time intervals. This condition makes every displayed integral finite and permits differentiation under the integral. Its validity for the original class (1.12)–(1.16) is not asserted. The conditional identities below assert no existence of positive-energy solutions in this auxiliary decay class. They provide neither a local construction nor a non-vacuity assertion for that class.

Integrations on $\mathbb{R}^2$ can then be justified with χ_L from (2.3). Every smooth rapidly decaying vector field F used below satisfies

$$\begin{aligned} \int \chi_L \operatorname{div} F \, dx &= - \int F \cdot \nabla \chi_L \, dx, \\ \left| \int F \cdot \nabla \chi_L \, dx \right| &\leq \frac{\|\nabla \eta\|_\infty}{L} \int_{\{|x| \geq L\}} |F| \, dx \longrightarrow 0. \end{aligned} \quad (3.1)$$

Dominated convergence treats the volume integrals. For integration by parts of two factors, apply this estimate to their product flux. Uniform decay supplies time-local dominating functions. These observations justify the boundary and time operations used below.

Write

$$d = \operatorname{div} u, \quad k = \frac{1}{2}|u|^2, \quad S = 2\mu D(u) + \lambda dI, \quad v = 2\mu + \lambda > 0. \quad (3.2)$$

Then $\operatorname{div} S = \mu \Delta u + (\mu + \lambda) \nabla d$ and $Q = S : \nabla u$. In two dimensions,

$$Q = 2\mu |D(u) - \frac{1}{2}dI|^2 + (\mu + \lambda)d^2 \geq 0. \quad (3.3)$$

Define

$$K(t) = \int \rho k \, dx, \quad H(t) = c_v \int \rho \theta \, dx, \quad E(t) = K(t) + H(t), \quad (3.4)$$

$$\mathcal{D}(t) = \int \{\mu |\nabla u|^2 + (\mu + \lambda)d^2\} \, dx. \quad (3.5)$$

Proposition 3.1 *Under the regularity condition above,*

$$K' + \mathcal{D} = \int P d \, dx, \quad H' = \mathcal{D} - \int P d \, dx, \quad E' = 0. \quad (3.6)$$

If the interval includes zero, with the stated initial traces, then

$$K(t) + H(t) = E_0, \quad \|\sqrt{\rho}u\|_2^2 \leq 2E_0, \quad \|P\|_1 \leq \frac{R}{c_v} E_0. \quad (3.7)$$

For any two times $s < t$ in the interval,

$$K(t) + \frac{\mu}{2} \int_s^t \|\nabla u\|_2^2 \, dt \leq K(s) + \frac{1}{\mu} \int_s^t \|P\|_2^2 \, dt. \quad (3.8)$$

Proof Continuity gives the transport identity

$$\partial_t(\rho f) + \operatorname{div}(\rho u f) = \rho \dot{f}. \quad (3.9)$$

The momentum equation is $\rho \dot{u} = \operatorname{div} S - \nabla P$. Dotting it with u , and using the symmetry of S , gives

$$\rho \dot{k} = \operatorname{div}(Su - Pu) - Q + Pd. \quad (3.10)$$

Integrate (3.9) with $f = k$ and apply (3.1) to obtain $K' + \int Q \, dx = \int P d \, dx$. Integration by parts twice gives

$$\int \partial_i u_j \partial_j u_i \, dx = - \int u_j \partial_j (\operatorname{div} u) \, dx = \int d^2 \, dx,$$

where repeated indices are summed over 1, 2. Since $2|D(u)|^2 = |\nabla u|^2 + \sum_{i,j} \partial_i u_j \partial_j u_i$,

$$\int Q dx = \mu \|\nabla u\|_2^2 + (\mu + \lambda) \|d\|_2^2 = \mathcal{D}. \quad (3.11)$$

The heat equation, (3.9) with $f = \theta$, and (3.1) with $F = \nabla \theta$ give $H' = \int Q dx - \int P d dx$. This proves (3.6). Add the first two identities, integrate in time, and use $\rho, \theta \geq 0$ to obtain (3.7).

Finally, $|d| \leq \sqrt{2}|\nabla u|$, and Cauchy–Schwarz followed by Young’s inequality gives

$$\left| \int P d dx \right| \leq \sqrt{2} \|P\|_2 \|\nabla u\|_2 \leq \frac{\mu}{2} \|\nabla u\|_2^2 + \frac{1}{\mu} \|P\|_2^2.$$

Since $\mathcal{D} \geq \mu \|\nabla u\|_2^2$, integration of the kinetic identity proves (3.8). $\square$

The physical energy conserves the sum of kinetic and internal energy. The term $\mathcal{D}$ transfers energy between them; it does not appear with a positive sign in the equation for E . Thus (3.7) supplies no bound on $\int_s^t \mathcal{D} dt$ by itself. If $M(t) = \|\rho(t)\|_\infty$,

$$\|P\|_2^2 \leq R^2 M(t) \int \rho \theta^2 dx, \quad (3.12)$$

but this introduces a higher thermal quantity and a density coefficient, neither bounded uniformly up to $T_{\max}$ by (3.7).

The effective viscous flux $G = \nu d - P$ describes the same coupling.

Proposition 3.2 *Under the same regularity condition,*

$$H' + \frac{1}{\nu} \|P\|_2^2 = \mathcal{D} - \frac{1}{\nu} \int P G dx, \quad (3.13)$$

$$K' + \frac{1}{2} \mathcal{D} \leq \frac{1}{2\nu} \|P\|_2^2. \quad (3.14)$$

Consequently,

$$\frac{1}{2\nu} \int_s^t \|P\|_2^2 dt \leq H(s) + \int_s^t \mathcal{D} dt + \frac{1}{2\nu} \int_s^t \|G\|_2^2 dt. \quad (3.15)$$

Proof Substitute $d = (G + P)/\nu$ in the internal-energy identity to obtain (3.13). The identity

$$\|\nabla u\|_2^2 = \|d\|_2^2 + \|\omega\|_2^2, \quad \omega = \partial_1 u_2 - \partial_2 u_1,$$

follows by expanding the squares and using $\int \partial_1 u_1 \partial_2 u_2 dx = \int \partial_2 u_1 \partial_1 u_2 dx$. Hence $\mathcal{D} = \nu \|d\|_2^2 + \mu \|\omega\|_2^2 \geq \nu \|d\|_2^2$, and

$$\int P d dx \leq \frac{1}{2\nu} \|P\|_2^2 + \frac{\nu}{2} \|d\|_2^2 \leq \frac{1}{2\nu} \|P\|_2^2 + \frac{1}{2} \mathcal{D}.$$

This proves (3.14). Finally, $-\int P G dx \leq \frac{1}{2} \|P\|_2^2 + \frac{1}{2} \|G\|_2^2$. Insert this in (3.13), integrate, and discard the nonnegative $H(t)$ to obtain (3.15). $\square$

Integration of (3.14) yields

$$\int_s^t \mathcal{D} dt \leq 2K(s) + \frac{1}{\nu} \int_s^t \|P\|_2^2 dt.$$

Inserting this in (3.15) leaves the coefficient $1/\nu$ of the pressure integral on the right, exceeding its coefficient $1/(2\nu)$ on the left. The two estimates therefore give no absorption. At the level of exact identities their sum is $E' = 0$. A further estimate independent of this exchange is needed.

4 Thermal sources and the pressure equation

Set $\beta = R/c_v$. Since $\dot{p} = -\rho d$ and $c_v \rho \dot{\theta} = \kappa \Delta \theta + Q - Pd$, the pressure equation is

$$\dot{P} + (1 + \beta)Pd = \beta(Q + \kappa \Delta \theta). \quad (4.1)$$

No division by density is involved. The two thermal sources cannot be discarded in favor of a source-free barotropic pressure equation.

Proposition 4.1 *Under the regularity condition of Section 3,*

$$\begin{aligned} \frac{d}{dt} \|P\|_2^2 + \frac{1+2\beta}{\nu} \int P^3 dx &= -\frac{1+2\beta}{\nu} \int P^2 G dx + 2\beta \int PQ dx \\ &\quad + 2\beta \kappa \int P \Delta \theta dx, \end{aligned} \quad (4.2)$$

$$\int P \Delta \theta dx = -R \int \rho |\nabla \theta|^2 dx - R \int \theta \nabla \rho \cdot \nabla \theta dx. \quad (4.3)$$

Proof Multiply (4.1) by $2P$ and use $\int u \cdot \nabla(P^2) dx = -\int dP^2 dx$. Then

$$\frac{d}{dt} \|P\|_2^2 + (1+2\beta) \int P^2 d dx = 2\beta \int PQ dx + 2\beta \kappa \int P \Delta \theta dx.$$

Substitute $d = (G + P)/\nu$ to obtain (4.2). Integration by parts with $\nabla P = R\rho \nabla \theta + R\theta \nabla \rho$ proves (4.3). All product fluxes satisfy (3.1). $\square$

The favorable cubic term in (4.2) is accompanied by $\int PQ dx \geq 0$ on the right. The last integral in (4.3) has no fixed sign. Neither term is controlled by (3.7).

For the thermal estimate, write

$$\begin{aligned} X &= \int \rho \theta^2 dx, \quad Z = \|\nabla \theta\|_2^2, \quad J = \int \theta Q dx, \\ V &= \int \rho |\dot{u}|^2 dx, \quad N = \|\nabla u\|_2^2. \end{aligned} \quad (4.4)$$

Proposition 4.2 *Under the regularity condition of Section 3,*

$$\frac{c_v}{2} X' + \kappa Z = J - R \int \rho \theta^2 d dx. \quad (4.5)$$

Writing $U = \|u\|_\infty$ and $M = \|\rho\|_\infty$, for every $\varepsilon > 0$ one has

$$J \leq \varepsilon(V + Z) + C_\varepsilon U^2 \{N + (1 + M)X\} + C\|d\|_\infty X. \quad (4.6)$$

Proof Multiply the heat equation by θ and apply (3.9) to $\theta^2/2$. Integration by parts of conduction gives (4.5). Testing the momentum equation by $u\theta$ gives

$$\begin{aligned} \int \theta \{ \mu |\nabla u|^2 + (\mu + \lambda) d^2 \} dx = & - \int \rho \theta u \cdot \dot{u} dx + \int P \theta d dx + \int P u \cdot \nabla \theta dx \\ & - \mu \int \partial_i u_j u_j \partial_i \theta dx - (\mu + \lambda) \int d u \cdot \nabla \theta dx. \end{aligned} \quad (4.7)$$

Because $\theta \geq 0$ and $\mu + \lambda \geq 0$, the left side is at least $\mu \int \theta |\nabla u|^2 dx$. Also, $0 \leq Q \leq (2\mu + 2|\lambda|) |\nabla u|^2$. Thus J is bounded by a constant times the sum of the absolute values on the right of (4.7). Cauchy–Schwarz gives

$$\begin{aligned} \left| \int \rho \theta u \cdot \dot{u} dx \right| & \leq U X^{1/2} V^{1/2}, \\ \left| \int P \theta d dx \right| & \leq R \|d\|_{\infty} X, \\ \left| \int P u \cdot \nabla \theta dx \right| & \leq R U (M X)^{1/2} Z^{1/2}. \end{aligned}$$

The two viscous cross terms are bounded by $CUN^{1/2}Z^{1/2}$, using $|d| \leq \sqrt{2}|\nabla u|$. Young's inequality gives

$$CUX^{1/2}V^{1/2} \leq \varepsilon V + C_\varepsilon U^2 X$$

and, choosing the two parameters with sum ε ,

$$CU(MX)^{1/2}Z^{1/2} + CUN^{1/2}Z^{1/2} \leq \varepsilon Z + C_\varepsilon U^2(MX + N).$$

Together these inequalities prove (4.6). □

With $d_- = \max\{-d, 0\}$,

$$-R \int \rho \theta^2 d dx \leq R \|d_-\|_{\infty} X. \quad (4.8)$$

Using this coefficient in Gronwall's inequality requires its time integrability, already part of (1.17). The quantities V , U , and M in (4.6) also remain to be controlled uniformly up to a finite $T_{\max}$.

For a time-independent $h \in C_c^\infty(\mathbb{R}^2)$ the localized temperature test is

$$\begin{aligned} \frac{c_v}{2} \frac{d}{dt} \int h \rho \theta^2 dx + \kappa \int h |\nabla \theta|^2 dx = & \int h \theta Q dx - \int h P \theta d dx \\ & + \frac{c_v}{2} \int \rho \theta^2 u \cdot \nabla h dx + \frac{\kappa}{2} \int \theta^2 \Delta h dx. \end{aligned} \quad (4.9)$$

Indeed, apply (3.9) to $\theta^2/2$ and multiply by h ; for conduction use

$$\kappa \int h \theta \Delta \theta dx = -\kappa \int h |\nabla \theta|^2 dx + \frac{\kappa}{2} \int \theta^2 \Delta h dx.$$

An extension to a growing weight requires control of both additional terms in (4.9). The density-weighted physical energy gives no unweighted θ^2 control in a vacuum region.

5 A total specific-energy multiplier

Pressure work and viscous heating can be cancelled before estimating them. Set

$$w = c_v \theta + k, \quad F = Su - Pu, \quad a_\kappa = \frac{\kappa}{c_v}, \quad Y = \int \rho w^2 dx. \quad (5.1)$$

Proposition 5.1 *Under the regularity condition of Section 3,*

$$\rho \dot{w} = \kappa \Delta \theta + \operatorname{div} F. \quad (5.2)$$

The functional Y satisfies

$$\frac{1}{2} Y' + \kappa c_v \|\nabla \theta\|_2^2 = -\kappa \int \nabla k \cdot \nabla \theta dx - \int F \cdot (c_v \nabla \theta + \nabla k) dx, \quad (5.3)$$

$$\frac{1}{2} Y' + a_\kappa \|\nabla w\|_2^2 = \int (a_\kappa \nabla k - F) \cdot \nabla w dx. \quad (5.4)$$

In particular,

$$\begin{aligned} Y' + a_\kappa \|\nabla w\|_2^2 &\leq a_\kappa^{-1} \|a_\kappa \nabla k - F\|_2^2 \\ &\leq C \int |u|^2 (|\nabla u|^2 + P^2) dx. \end{aligned} \quad (5.5)$$

With N as in (4.4), $U = \|u\|_\infty$, and $M = \|\rho\|_\infty$,

$$\begin{aligned} Y(t) + a_\kappa \int_s^t \|\nabla w\|_2^2 dt &\leq \left(Y(s) + C \int_s^t U^2 N dt \right) \\ &\quad \times \exp \left(C \int_s^t M U^2 dt \right). \end{aligned} \quad (5.6)$$

Proof Add (3.10) to the material form of the heat equation. The terms $Q - Pd$ cancel and give (5.2) without division by density. Since $Su = \mu \nabla k + \mu(u \cdot \nabla)u + \lambda ud$, equivalently

$$\rho \dot{w} = \kappa \Delta \theta + \mu \Delta k + \mu \operatorname{div}((u \cdot \nabla)u) + \lambda \operatorname{div}(ud) - \operatorname{div}(Pu).$$

Apply (3.9) to $w^2/2$ and test (5.2) by w . Then

$$\frac{1}{2} Y' = -\kappa \int \nabla w \cdot \nabla \theta dx - \int F \cdot \nabla w dx. \quad (5.7)$$

The substitutions $\nabla w = c_v \nabla \theta + \nabla k$ and $\nabla \theta = c_v^{-1}(\nabla w - \nabla k)$ respectively prove (5.3) and (5.4).

Cauchy–Schwarz and Young’s inequality yield

$$\left| \int (a_\kappa \nabla k - F) \cdot \nabla w dx \right| \leq \frac{a_\kappa}{2} \|\nabla w\|_2^2 + \frac{1}{2a_\kappa} \|a_\kappa \nabla k - F\|_2^2.$$

Insert this in (5.4) and multiply by two. The pointwise bounds $|\nabla k| \leq |u| |\nabla u|$ and $|S| \leq C |\nabla u|$ imply

$$|a_\kappa \nabla k - F|^2 \leq C |u|^2 (|\nabla u|^2 + P^2),$$

which proves (5.5). Positivity also gives

$$Y \geq c_v^2 X + \frac{1}{4} \int \rho |u|^4 dx, \quad \|P\|_2^2 \leq \frac{R^2}{c_v^2} MY. \quad (5.8)$$

Consequently,

$$Y' + a_\kappa \|\nabla w\|_2^2 \leq CU^2 N + CMU^2 Y. \quad (5.9)$$

Multiply by $\exp(-C \int_s^t MU^2 d\tau)$ and integrate. After multiplying back, the dissipation integral has the weight $\exp(C \int_\tau^t MU^2 d\sigma) \geq 1$. The forcing term is at most $\exp(C \int_s^t MU^2 d\tau) \int_s^t CU^2 N d\tau$. Dropping the excess dissipation weight proves (5.6). $\square$

The positive functional Y contains both the density-weighted temperature square and velocity fourth power. However, $\nabla w = c_v \nabla \theta + \nabla k$ controls only their sum. The identity (5.3) retains the mixed conduction term, whereas (5.4) places the other terms in a flux. No small factor depending on E_0 has been established on the right of (5.5). The coefficients in (5.6) have not been bounded uniformly at a finite maximal time under the original data assumptions.

Small $\int \rho_0 w_0 dx = E_0$ does not imply small $\int \rho_0 w_0^2 dx$. To see the distinction, choose a smooth nonnegative compactly supported density equal to one near the origin and a nonnegative nonzero $\psi \in C_c^\infty(\mathbb{R}^2)$. For $r > 0$ sufficiently small, set

$$u_0 = 0, \quad \theta_0(x) = b\psi(x/r), \quad b > 0,$$

with the thermal support inside the region where $\rho_0 = 1$. The data satisfy (1.8)–(1.9). Condition (1.10) holds with $g = R\nabla \theta_0$, extended by zero away from its support. A change of variables gives

$$\begin{aligned} E_0 &= c_v b r^2 \int \psi dx, & \|P_0\|_2^2 &= R^2 b^2 r^2 \int \psi^2 dx, \\ Y(0) &= c_v^2 b^2 r^2 \int \psi^2 dx. \end{aligned} \quad (5.10)$$

For fixed positive small E_0 , take $r^2 = E_0 / (c_v b \int \psi dx)$ and let b increase. Both $\|P_0\|_2^2$ and $Y(0)$ then grow linearly in b . The higher initial norms are finite for each datum but not uniform in this family. This rules out an instantaneous bound using E_0 alone; it does not rule out a spacetime estimate using smoothing and prescribed higher initial bounds, and gives no conclusion against global existence.

6 Remaining estimates and the continuation question

Theorem 2.2 is complete under its L^1 forcing and L^2 flux assumptions. The energy and thermal estimates are proved under the explicit smoothness and decay condition of Section 3. Their use for every solution in (1.12)–(1.16) requires the corresponding cutoff limits, nonlinear tests, and initial traces. For example, the w^2 test requires integrability of $\rho |u|^4$ and its flux terms. A formal multiplier or the weak far-field phrase does not prove these facts.

Even after those justifications, a chain of unresolved estimates remains. The kinetic inequality (3.8) requires the pressure-square time integral. The effective-flux identities (3.13)–(3.15) do not provide an absorbable bound for it. The pressure equation introduces $\int PQ dx$ and the mixed density-gradient term in (4.3). The thermal test requires temperature-weighted heating and a compression coefficient. The total specific-energy functional replaces the explicit heating and pressure work by $\int |u|^2 (|\nabla u|^2 + P^2) dx$, which remains uncontrolled by the physical energy estimate.

Continuation requires constants remaining finite as $T \uparrow T_{\max} < \infty$, sufficient to prove

$$\sup_{T < T_{\max}} \int_0^T (\|\operatorname{div} u\|_{\infty} + \|u\|_{4/\alpha}) dt < \infty.$$

Assuming this integral finite to estimate the preceding coefficients cannot establish its finiteness from small initial energy. A bootstrap density bound or velocity supremum bound must likewise be recovered from the equations with uniform constants.

The results therefore identify positive cancellations and exact requirements for a future closure. They establish neither a small-energy global-existence theorem in the original local class nor a contradiction to such a theorem. An independent estimate connecting the pressure and thermal flux terms to the continuation quantities is still needed, without adding assumptions to the target data.

References

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